

Masoud (Mas) Hakimi Heris

CONTACT INFORMATION

Ph.D. Candidate, Electrical Engineering

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RESEARCH INTERESTS

I work on hardware–algorithm co-design for early fault-tolerant quantum computation. Algorithms, architectures, and physical control are usually developed in separate layers joined by worst-case abstractions. I design them together, around the native resources and error mechanisms of a specific device. In practice this means tracing an algorithm’s accuracy requirement down to a phase-angle tolerance, then to a compiled gate requirement, then to a pulse-level control constraint, and optimizing the whole chain at once.

My interests span quantum algorithms and QSP/QSVT, quantum error analysis and fault tolerance, adaptive quantum control, hybrid CV–DV computation, photonic quantum computing, cavity QED, and spin–photon interfaces.

HIGHLIGHTS

- **Record:** 7 peer-reviewed publications, 3 preprints or manuscripts under review, and 4 in preparation; 5 of those 7 in-progress works are first- or co-first-authored. 90 citations, *h*-index 3 (Google Scholar, August 2026). Venues include IEEE QCE, HPCA (under review), SPIE Photonics for Quantum, and *APL Computational Physics*.
- **Architecture:** Designed a native trapped-ion qudit–oscillator instruction set and proved three of its primitives generate universal Hamiltonians; compiling dissipative vibronic dynamics onto a single $^{171}\text{Yb}^+$ ququart cut gate count from 96 to 28 per Trotter step against the superconducting realization.
- **Control:** Built HyPulse, a GRAPE-based pulse-synthesis framework for hybrid CV–DV gates, including single-step Toffoli synthesis from native trapped-ion interactions.
- **Algorithms:** Showed QSP block-encoding and phase-angle errors cross-compensate to first order, and built a constrained optimizer suppressing correlated control error by $4\text{--}19\times$; co-led the first experimental realization of QSP on a programmable photonic integrated circuit.
- **Funding:** Co-Entrepreneurial Lead on a \$50K NSF I-Corps Teams (Quantum+X) award; supported as GRA on NSF NQVL QACTI (Duke subcontract).
- **Teaching and mentoring:** Preparing the Professoriate Fellow, co-teaching ECE 211 (Electric Circuits); mentored two undergraduate researchers on machine-learning approaches to quantum control.
- **Service:** Sub-reviewer for 5 manuscripts across *IEEE J-STSP*, IEEE QCE, and the SFWM workshop at SC; K–12 outreach volunteer at Montana State; member, IEEE and IEEE Photonics Society.

EDUCATION

Ph.D. Candidate, Electrical Engineering, North Carolina State University

2024–Present

Advisor: Prof. Yuan Liu · Expected May 2027 · GPA: 4.0/4.0

Dissertation: *Co-Design of Quantum Algorithms, Architectures, and Control for Early FT Hardware*

M.S., Electrical Engineering, Montana State University

2022–2024

Advisor: Prof. David L. Dickensheets · GPA: 3.87/4.0

Thesis: *MEMS Cantilevers for Dynamic Strain Studies of 2D Materials*

M.S., Electrical Engineering (Electronics), Qazvin Azad University, Iran

2012–2015

B.S., Electrical Engineering (Electronics), Karaj Azad University, Iran

2006–2011

JOURNAL ARTICLES

- [J3] E. Karooby, H. Sahbafar, **M. Hakimi Heris**, A. Hadi, V. Eskandari. “Identification of Low Concentrations of Flucytosine Drug Using a SERS-Active Filter Paper Substrate.” *Plasmonics* 19(2), 855–863, 2024.
- [J2] V. Eskandari, H. Sahbafar, E. Karooby, **M. Hakimi Heris**, S. Mehmandoust, D. Razmjoue, A. Hadi. “SERS Filter Paper Substrates Decorated with Silver Nanoparticles for the Detection of Molecular Vibrations of Acyclovir Drug.” *Spectrochimica Acta Part A* 298, 122762, 2023.
- [J1] K. J. Joven, E. R. Das, J. Bierman, A. Majumdar, **M. Hakimi Heris**, Y. Liu. “Scalable Quantum Computational Science: A Perspective from Block-Encodings and Polynomial Transformations.” *APL Computational Physics* 2, 010901, 2026.

CONFERENCE PAPERS

- [C4] **M. Hakimi Heris**, Y. Liu, F. Mueller. “HyPulse: A Pulse Synthesis Framework for Hybrid Qubit–Oscillator Gates on Trapped-Ion Platform.” *IEEE International Conference on Quantum Computing and Engineering (QCE)*, 2026. arXiv:2604.26804.
- [C3] E. Karooby, **M. Hakimi Heris**, Q. Gu, D. Farfurnik. “Inverse Design of Non-Periodic Circular Cavities for Efficient Spin–Photon Interfacing by Genetic Algorithm.” *SPIE Optical Modeling and Performance Predictions XV* 13594, 116–119, 2025.
- [C2] A. Das, N. K. Vij, **M. Hakimi Heris**, D. Farfurnik. “Optimization of Circular Gratings via Guided-Mode-Expansion-Method-Based Inverse Design.” *SPIE Photonics for Quantum* 13563, 65–71, 2025.
- [C1] **M. Hakimi Heris**, R. Faez. “Investigation of the Possibility of Using Single-Electron Transistors in Digital-to-Time Converters.” *IEEE Global Congress on Electrical Engineering (GC-ElecEng)*, 150–153, 2021.

- [U3] C. Chu, **M. Hakimi Heris**, H. Zhou, Y. Liu. “FACT: A Fidelity-Aware Compiler for Trapped-Ion Quantum Computers.” Under review, *IEEE International Symposium on High-Performance Computer Architecture (HPCA)*, 2027.
- [U2] R. Patel, **M. Hakimi Heris**, Y. Liu, F. Mueller. “Synthesizing Compound Pulse Gadgets for Hamiltonian Simulation on Trapped-Ion Platforms.” arXiv:2607.00826, 2026.
- [U1] **M. Hakimi Heris**. “Graph State Generation Based on Quantum Photonic Devices, Enabling Measurement-Based Quantum Computing.” arXiv:2607.16667, 2026.

- [P4] E. Karooby*, **M. Hakimi Heris***, Y. Liu, Q. Gu. “Universal Function Approximation Using Quantum Signal Processing on a Photonic Integrated Platform.” * Equal contribution.
- [P3] **M. Hakimi Heris**, Y. Liu. “Robustness-Aware Quantum Signal Processing: Error Cancellation, Optimization, and Learned Angle Synthesis.”
- [P2] **M. Hakimi Heris**, Y. Liu, F. Mueller. “A Native Trapped-Ion Hybrid Qudit–Oscillator Instruction Set.”
- [P1] **M. Hakimi Heris**, Y. Liu, F. Mueller. “Learned Pulse Synthesis for Hybrid CV–DV Gates on Trapped-Ion Hardware.”

Grand Unification of Quantum Algorithms on Photonic Integrated Circuits

- Source of Support: NSF I-Corps Teams (Quantum+X)
- Role: Co-Entrepreneurial Lead
- Project Dates: 2026
- PI: Qing Gu (NC State, ECE/Physics); Technical Co-Lead: Yuan Liu (NC State, ECE/CS); Co-Entrepreneurial Lead: Elaheh Karooby; Industry Mentor: Amirhassan Shams-Ansari (Monarch Quantum)
- Total Project Amount: \$50,000

NQVL:QTSD:Pilot/Design: Quantum Advantage-Class Trapped Ion System (QACTI)

- Source of Support: NSF National Quantum Virtual Laboratory, via subcontract from Duke University
- Role: Supported as Graduate Research Assistant
- Lead PI: Ken Brown (Duke University); NC State: Yuan Liu, Frank Mueller
- Phase 1 (NSF 2410675): \$100,000, 08/2024–07/2025
- Phase 2 (NSF 2531350): \$390,000, 09/2025–08/2027
- Building an advantage-class trapped-ion quantum computer for remote use by the scientific community, with an open-source hardware/software stack.

- “Experimental Realization of Quantum Signal Processing on a Photonic Integrated Circuit,” IEEE Photonics Conference. *Poster.* Nov. 2026
- “HyPulse: A Pulse Synthesis Framework for Hybrid Qubit–Oscillator Gates on Trapped-Ion Platform,” IEEE International Conference on Quantum Computing and Engineering (QCE), Toronto, Canada. *Contributed talk.* Sept. 2026
- “Toward Compact Multiqubit Gates on Trapped Ions: A Hardware-Aware Pulse Synthesis,” NSF Workshop on Signal and Information Processing in the Quantum Era (SIPQ), IEEE Quantum Week/QCE, Toronto, Canada. *Poster.* Sept. 2026
- “Toward Compact Multiqubit Gates on Trapped Ions: A Hardware-Aware Pulse Synthesis,” QCCC-26: Fifth International Workshop on Quantum Classical Cooperative Computing, co-located with ISCA. *Contributed talk.* June 2026
- “Synthesizing Compound Pulse Gadgets for Hamiltonian Simulation on Trapped-Ion Platforms,” QCCC-26: Fifth International Workshop on Quantum Classical Cooperative Computing, co-located with ISCA. *Contributed talk.* June 2026
- “Simulations of Nonadiabatic Vibronic Dynamics on Hybrid Qudit–Qumode Trapped-Ion Devices,” RQS Institute Workshop, University of Maryland. *Poster.* June 2026
- “MEMS Devices for In-Situ Straining of 2D Materials,” MonArk NSF Quantum Foundry Site Visit, Bozeman, MT. 2024
- “MEMS Cantilevers for Dynamic Strain Studies of 2D Materials,” Optical Science & Engineering Conference (OpTeC), Bozeman, MT. 2023

AWARDS AND
PROFESSIONAL
PROGRAMS

Prospects in Theoretical Physics (PiTP), Institute for Advanced Study, Princeton	2026
Graduate Merit Award, College of Engineering, NC State University	2024–2025
AIM Photonics Summer Academy, MIT	2023
Summer of Quantum Short Course, LPS Qubit Collaboratory	2023
Excellent Graduate Student in Electronics Engineering, Qazvin Azad University	2015
Top 3 Undergraduate Student in Electronics Engineering, Karaj Azad University	2009
Best Patent, Kharazmi Festival, Tehran, Iran	2004

RESEARCH
EXPERIENCE

Graduate Researcher, Quantum Computing and Engineering, NC State University 2024–Present
Advisor: Prof. Yuan Liu

- Designed a native trapped-ion hybrid qudit–oscillator instruction set and proved that three of its primitives generate universal Hamiltonians, with no synthesized mode-coupling operation required.
- Compiled dissipative vibronic dynamics of a three-chromophore complex onto a single $^{171}\text{Yb}^+$ ququart, cutting gate count from 96 to 28 per Trotter step relative to the superconducting realization and reproducing published trajectories to $\sim 1.1\%$ RMSE.
- Built HyPulse, a GRAPE-based pulse-synthesis framework for hybrid CV–DV gates, and extended HybridLane (PennyLane) to native qudit–oscillator circuits.
- Co-led the first experimental realization of QSP on a programmable photonic integrated circuit, synthesizing nonlinear functions on linear-optical hardware.
- Optical characterization and design of nanophotonic structures for light–matter interaction (cavity QED).
- Collaborations across ECE, physics, and computer science, including Duke University’s NQVL for trapped-ion experimental validation.

Graduate Researcher, MonArk NSF Quantum Foundry, Montana State University 2022–2024
Advisors: Prof. David L. Dickensheets, Prof. Nicholas J. Borys

- Designed, fabricated, and characterized MEMS/NEMS devices for optical studies of 2D quantum materials, from EBL and photolithography through SEM/AFM metrology.

TEACHING
EXPERIENCE

Preparing the Professoriate (PTP) Fellow, NC State University 2026–2027
Faculty mentor: Dr. Jennifer Marley (ECE). Year-long teaching apprenticeship: co-teaching **ECE 211, Electric Circuits**, the required sophomore circuits course for EE, CPE, and BME majors, while developing a teaching portfolio and philosophy statement and completing pedagogy coursework.

Teaching Assistant, Circuits and Systems Laboratories 2022–2025
NC State University, Signals, Circuits, and Systems Laboratory (~ 25 students per section). Montana State University, Electrical Circuits Laboratory (~ 20 students per section, 2022–2023). Instructed students in building and characterizing RLC and power circuits, bench measurement, and data analysis, with an emphasis on clear technical reporting and experimental documentation.

Teaching Assistant, Microfabrication Laboratory, Montana State University 2022–2023
 ~ 20 students per section. Prepared wafers, tooling, and process recipes ahead of each session, then walked students through a full multi-layer transistor fabrication flow in a cleanroom: oxidation, photolithography, deposition, and both wet and dry etching.

COURSES PREPARED TO TEACH

Core undergraduate: circuits, signals and systems, electronics, semiconductor devices, microfabrication and nanofabrication laboratory.
Graduate and elective: quantum computing architectures, quantum control and pulse-level programming, quantum algorithms and QSP/QSVT, photonic and quantum devices.

MENTORING

- Samuel Firmansyah, Undergraduate, Computer Science, NC State University

Project: AI-based robust quantum signal processing (learned robust angle synthesis).

Spring 2026
- Nidhi Grover, Undergraduate, Computer Science, NC State University

Project: AI-based pulse synthesis for hybrid CV-DV gates on trapped-ion hardware.

Spring 2026

PROFESSIONAL
SERVICE

PEER REVIEW

- Sub-reviewer on program-committee and editorial assignments for Prof. Yuan Liu and Prof. Frank Mueller (5 manuscripts, 2025–2026):
- *IEEE Journal of Selected Topics in Signal Processing*, Special Issue on Quantum Technologies, 1 manuscript (2026)
 - IEEE International Conference on Quantum Computing and Engineering (QCE, IEEE Quantum Week), 2 manuscripts in the Quantum Applications (QAPP) and Quantum Machine Learning tracks (2026)
 - International Workshop on Software Frameworks and Workload Management on Quantum and HPC Ecosystems (SFWM) at SC, 2 manuscripts (2025, 2026)

MEMBERSHIPS

- Student Member, IEEE

2026–Present
- Member, IEEE Photonics Society

2026–Present

OUTREACH

- K–12 Outreach Volunteer, Montana State University

2023–2024
- Ran hands-on demonstrations for visiting school groups at MSU’s annual campus science outreach day, alongside research groups from across the university: simple photonics experiments, plus a cleanroom station where students tried on fabrication garments and saw how devices are made.

TECHNICAL
SKILLS

- Quantum algorithms & theory:** QSP/QSVT, block encodings, Hamiltonian simulation, open quantum systems, hybrid CV-DV computation, quantum error analysis.
- Quantum control:** GRAPE, pulse synthesis, constrained optimization, differentiable simulation, hardware-aware compilation, Hamiltonian-based control.
- Software:** Python, JAX, QuTiP, PennyLane, Qiskit, Strawberry Fields, MATLAB, Julia, C/C++.
- Machine learning:** Neural networks, Fourier neural operators, learned pulse synthesis, physics-informed and differentiable training.
- Photonics & hardware:** Programmable photonic circuits, Lumerical FDTD/INTERCONNECT/qINTERCONNECT, COMSOL, nanophotonic design and fabrication, optical characterization.

EARLIER INDUSTRY
EXPERIENCE

- Software and electrical engineering roles, 2011–2021 (FDM4 Canada; power and industrial systems engineering and project management).